🏆 SOLARIS X HACKATHON 2026 — OFFICIAL SUBMISSION

PersonalOS Agent

An autonomous, voice-controlled multi-agent AI that monitors your Gmail and Google Calendar, reasons about what actions to take, calls your phone for approval — and executes on your behalf. **No buttons. Just speak.**

- 3 autonomous agents (Observer · Planner · Executor)
- Voice approval via Twilio — fully working
- Real Gmail + Google Calendar integration
- MCP · Redis · ChromaDB · FastAPI

1 · What Is PersonalOS Agent?

PersonalOS Agent is a **production-grade agentic pipeline** that acts as your personal AI secretary. It watches your inbox and calendar 24/7, thinks about what needs to happen, and either acts automatically or calls your phone and asks — in plain speech. You reply by talking. No app to open, no buttons to press.

3	6	33	7
AUTONOMOUS AGENTS	MCP TOOLS	TESTS PASSING	MEETING SCENARIOS

💡 **The core idea:** Most AI assistants wait for you to ask them something. PersonalOS Agent *proactively* monitors your digital life, makes decisions autonomously, and only interrupts you when genuinely needed — via a phone call, not a notification.

2 · System Architecture

```
┌──────────────────────────────────────────────────────────────────────────────────────────────────────┐ │ PersonalOS Agent │ │ │ │ ┌──────────────┐
events:queue ┌──────────────┐ approvals:pending │ │ │ OBSERVER │ ──────────────────▶ │ PLANNER │ ──────────────────────▶ │ │ │
(The Eyes) │ (Redis) │ (The Brain) │ (Redis) │ │ └──────────┬──────────┘ └──────────────┘ │ │ │ │ LLM via OpenRouter │ │ │ │
ChromaDB preferences │ │ │ ▼ │ │ │ ┌──────────────────┐ │ │ │ │ EXECUTOR │ │ │ │ │ (The Hands) │ │ │ │
```

```
└──────┬──────┘ │ │ │ │ │ │ ┌──────────────────────────────────────────────┐ │ │ └──▶│ MCP Tool Server
:8000 (SSE JSON-RPC) │ │ │ │ read_emails · send_email · read_calendar │ │ │ create_event · list_files · move_file │ │ │
└──────────────────────────────────────────────┘ │ │ │ │ ┌──────────────┐ ┌──────────────┐
┌──────────────────┐ │ │ │ Dashboard :8080 │ │ Redis :6379 │ │ Twilio Voice API │ │ │ │ FastAPI + WS │ │ queues +
dedup│ │ 📞 calls your phone │ │ │ └──────────────┘ └──────────────┘ └──────────────────┘ │ │ │ │
┌──────────────────────────────────────────────────────────┐ │ │ │ ChromaDB (vector memory) │ │ │ │ user
preferences · historical approval rates (self-improving) │ │ │
└──────────────────────────────────────────────────────────┘ │
└──────────────────────────────────────────────────────────────┘
```

👁️ Observer Agent

Polls Gmail (unread emails), Google Calendar (next 3 days), and a sandboxed filesystem every 60 seconds via MCP tools. Deduplicates events across restarts using Redis SETNX. Detects urgency keywords for priority boosting. Can be triggered manually via dashboard.

🧠 Planner Agent

Consumes events from Redis queue. Calls OpenRouter LLM with structured JSON schema, urgency context, and ChromaDB preferences. Produces action plans with confidence 0–100. Applies decision scoring: `adjusted = base × urgency_mult × history_mult`

⚡ Executor Agent

Routes by confidence: >90% auto-executes + notifies, 70–89% → dashboard + voice call, <70% silently discards. Records every outcome in ChromaDB so future confidence scoring improves over time.

3 · Voice Approval — The "Wow" Feature

This is the feature that makes judges stop and pay attention. Instead of opening an app and clicking a button, you **pick up your phone and talk**. The AI understands what you say — approve, reject, or modify with a spoken instruction — and acts in real time.

```
📧 Email arrives ("reply to professor's urgent deadline email") ↓ Planner scores it: confidence = 78% → requires
voice approval ↓ Executor stores plan in Redis voice:plan:{id} with 5-min TTL ↓ Twilio places outbound call to your
phone url=/api/twilio/voice/{id} ↓ You answer: "Hey! Requested action: send email. Subject: Assignment Deadline. Say
yes to approve, no to reject, or modify followed by instructions." ↓ You say: "reply that I will submit it by 6pm
today, please give me some more time" ↓ Twilio POSTs SpeechResult to POST /api/twilio/speech/{id} ↓ LLM classifies
intent: MODIFY — instruction: "I will submit it by 6pm today, please give me some more time" ↓ plan["action_args"]
["body"] = voice instruction injected into email body plan["approved_override"] = True → re-queued to
```

```
approvals:pending ↓ Executor auto-executes: send_email(to, subject, body_with_voice_instruction) ↓ You hear: "Got it! I will send email with your changes. Goodbye!" ↓ 📊 Dashboard updates live via WebSocket broadcast
```

Intent Classification — 4 states

WHAT YOU SAY	INTENT	ACTION TAKEN
"Yes", "Sure", "Go ahead", "Do it"	APPROVE	Executes immediately, email sent
"No", "Don't", "Cancel", "Reject"	REJECT	Removed from dashboard, logged as rejected
"Modify …", "Reply that …", "But make it …"	MODIFY	Voice instruction injected into email body, then executed
Mumble / background noise / silence	UNCLEAR	Re-prompts you to speak clearly

Fallback safety: If LLM fails, word-boundary regex fallback takes over. "No" does not match "know" or "notify" — uses `\b` regex to prevent false triggers. `speechTimeout="10"` gives you 10 full seconds to speak. Say "done" or "stop" to end your instruction.

4 · Smart Meeting Intelligence

PersonalOS Agent now reads your calendar and cross-references it against incoming meeting request emails. It automatically drafts a polished reschedule or confirmation email based on what your calendar actually shows — no manual checking required.

```
📧 Email arrives: "Can we meet tomorrow at 3pm?" ↓ Observer detects meeting keywords → pushes event to queue ↓ Planner._build_prompts() calls read_calendar(days_ahead=3) via MCP ↓ MeetingResolver parses busy slots + finds free 1-hour windows Conflict check: Is tomorrow 3–4pm busy? ↓ YES → inject calendar context + free slots into LLM prompt ↓ LLM drafts polite reschedule email with specific alternate times ↓ Executor routes to voice approval → you approve by speaking → email sent
```

7 Scenarios Handled Automatically

1 Direct Conflict — same time requested as existing meeting — Conflict

Incoming email: "Can we meet tomorrow at 3pm?"

Calendar: Tomorrow 3–4pm → "Project Review"

AUTO-DRAFTED REPLY:

Hi, I already have something scheduled at 3pm tomorrow. Would 5pm tomorrow or any time on Friday work for you instead?

2 Entire Day Packed — no slots available — Fully Blocked

Incoming email: "Free anytime Thursday?"

Calendar: Thursday has back-to-back events 9am–6pm

AUTO-DRAFTED REPLY:

Thursday looks fully packed on my end. I'm available Friday morning from 10am — does that work for you?

3 Partial Overlap — meeting runs into existing event

Partial

Incoming email: "Let's do a 1-hour call at 2pm Friday"

Calendar: Friday 2:30–3:30pm → "Team Standup"

AUTO-DRAFTED REPLY:

I have a conflict from 2:30pm on Friday. Could we do 1pm instead (it ends before my 2:30 meeting), or push to Monday?

4 No Specific Time — open-ended availability request

Free Slots

Incoming email: "Let's catch up this week sometime"

Calendar: Wednesday afternoon and Friday morning are free

AUTO-DRAFTED REPLY:

Happy to catch up! I'm free Wednesday after 3pm or Friday before noon — let me know what works best for you.

5 Next Day Also Busy — cascading conflict

Multi-day

Incoming email: "Can we reschedule to tomorrow?"

Calendar: Tomorrow fully blocked AND day-after only morning free

AUTO-DRAFTED REPLY:

Tomorrow is fully booked as well. I do have Thursday morning free — say 10am or 11am?

6 Recurring Meeting Conflict — weekly block

Recurring

Incoming email: "Every Monday 2pm works?"

Calendar: Monday 2pm → recurring "Weekly Sync"

AUTO-DRAFTED REPLY:

Mondays at 2pm are blocked with a recurring meeting. Would Tuesday or Wednesday at the same time work as a regular slot?

7 Calendar Is Free — auto-create event + confirm

Auto-Schedule

Incoming email: "Can we meet tomorrow at 11am?"

Calendar: Tomorrow 11am–12pm is completely free

AUTO-DRAFTED REPLY + CALENDAR INVITE CREATED:

Great, I've blocked 11am–12pm tomorrow on my calendar. See you then! [Calendar invite sent]

5 · Complete Feature List

FEATURE	STATUS	TECHNICAL DETAIL
Real Gmail monitoring	✓ Live	OAuth 2.0, reads unread emails every 60s, deduplicates via Redis SETNX
Google Calendar integration	✓ Live	Reads events for next 7 days, creates events, same OAuth token as Gmail
LLM action planning	✓ Live	OpenRouter (any model), structured JSON schema, 2 alternatives per plan
Confidence-based routing	✓ Live	>90% auto-exec, 70–89% voice/dashboard, <70% discard with log
Voice approval via phone call	✓ Live	Twilio outbound call, TwiML Gather speech, 10s timeout, LLM + regex fallback
Voice modification	✓ Live	Say "modify [instruction]" — words injected into email body, then sent
Smart meeting scheduling	✓ Live	Calendar conflict detection, free slot finder, 7 scenarios, auto email draft
Human approval dashboard	✓ Live	FastAPI + vanilla JS, light/dark theme toggle, search, filter, bulk approve
Real-time WebSocket updates	✓ Live	Dashboard refreshes instantly on any plan change, 20s keep-alive ping
Vector memory (ChromaDB)	✓ Live	Stores user preferences + approval outcomes, influences future confidence scores
Decision scoring formula	✓ Live	adjusted = base × urgency_mult(1.0–1.3) × history_mult(0.9–1.1)
MCP tool server	✓ Live	FastMCP over SSE, agents never call Google APIs directly
Filesystem assistant	✓ Live	Sandboxed list + move files, directory traversal blocked at tool layer
Inject Event (demo mode)	✓ Live	Dashboard UI to push synthetic events without Google OAuth — perfect for live demos
Analytics dashboard	✓ Live	Confidence histogram, priority/response breakdown, queue depth, auto-exec rate
33 automated tests	✓ Passing	18 unit tests + 15 integration tests, covers full approve/reject lifecycle
Light/Dark theme toggle	✓ Live	CSS variables, persisted in localStorage
Structlog + JSON logging	✓ Live	Colorized console + structured JSON file for each agent

6 · Technology Stack

Python 3.11 | FastAPI | Redis | ChromaDB | OpenRouter LLM | MCP (FastMCP)
Twilio Voice API | Gmail API v1 | Google Calendar API v3 | OAuth 2.0 | WebSocket
sentence-transformers | asyncio | pydantic-settings | structlog | pytest (33 tests)

LAYER	TECHNOLOGY	WHY THIS CHOICE
Agents	Python asyncio tasks	Single event loop for all agents; no thread contention; clean CancelledError handling
Tool Protocol	MCP (Model Context Protocol)	Industry standard for AI tool access; agents are fully decoupled from API implementations
LLM	OpenRouter (OpenAI-compatible)	Swap between GPT-4o, Claude, Qwen, Llama without changing a line of code
Message queue	Redis BLPOP / BRPOP	Blocking pop = zero CPU when idle; atomic deduplication via SETNX
Vector memory	ChromaDB + all-MiniLM-L6-v2	Fast semantic search for preferences; persisted across restarts
Voice	Twilio TwiML + Gather speech	Enhanced ASR, 10s timeout, 10 speech recognition parameter options

7 · Live Demo Guide (for Judges)

Three paths to demo — choose based on available internet / OAuth setup.

🚀 Option A — Full Live Demo (most impressive)

- **Start the system**
  `python main.py` → all 3 agents + MCP server + dashboard boot in ~3 seconds
- **Open dashboard**
  Navigate to `http://localhost:8080` — health strip shows all services green
- **Inject a meeting request email**
  Click "Inject Event" tab → From: `professor@university.edu` → Subject: "Can we meet tomorrow at 3pm?" → check Urgent → click Inject
- **Watch it flow in real time**
  Activity feed updates live — Observer detected → Planner reasoned → calendar checked → Executor routing
- **Phone rings (if Twilio configured)**
  Answer and say **"Yes"** to approve, or **"Modify, reply that I'm available at 5pm instead"**
- **Email sent — dashboard updates**
  Email list shows "approved" badge. Activity log shows the voice instruction that was used.

📊 Option B — Seed Script Demo (no Google OAuth needed)

```
# Terminal 1
python main.py --skip-poll

# Terminal 2 — inject 4 realistic events instantly
python tests/fixtures/seed_events.py
```

🎙️ Option C — Voice-Only Demo

```
# Inject a voice approval test directly (calls your phone)
python scripts/test_voice_email.py

# Or trigger a test call from the dashboard header button
# Click "📞 Test Call" → phone rings → say anything to demonstrate ASR
```

Key things to show judges

- **The phone call itself** — this is the single most impressive thing. Have ngrok running and answer the call live.
- **The "modify" flow** — say a full sentence after "modify", show the exact text appears in the sent email body.
- **Dashboard real-time update** — after approving by voice, show the dashboard badge changes instantly (WebSocket).

- **Analytics tab** — show confidence distribution, auto-execute rate, this demonstrates the scoring system.
- **Meeting scenario** — inject a meeting request email, show it reads calendar and drafts a conflict email automatically.
- **ChromaDB memory** — Preferences tab shows stored user preferences that influence every planning decision.
- **Light/dark toggle** — small detail that shows polish.

8 · Why This Should Win

🏗️ PRODUCTION QUALITY

Not a prototype. 33 passing tests. Structured logging. Graceful shutdown. Pre-flight health checker. Retry logic with exponential backoff. Error surfaces to dashboard instead of silently dying.

📞 VOICE IS REAL

The phone call feature is fully working — confirmed by a real email arriving at vikas935314@gmail.com with the spoken voice instruction in the body. This is not a mock or simulation.

🤝 REAL INTEGRATIONS

Live Gmail, live Google Calendar, live Twilio. All authenticated with real OAuth. Not mock data — real emails flowing through the pipeline.

🧠 SELF-IMPROVING

ChromaDB records every approval/rejection. Future confidence scores adjust based on historical approval rates. The system gets smarter with every interaction.

🔌 MCP PROTOCOL

Uses the industry-standard Model Context Protocol — the same protocol Anthropic built into Claude. Agents never call external APIs directly, making the tool layer swappable and auditable.

💡 NOVEL UX

The interaction model is genuinely new: AI monitors → reasons → calls you → listens → acts. No app to open, no button to press. Your phone is the interface.

🏆 One-Line Pitch for Judges

PersonalOS Agent is an autonomous 3-agent AI system that watches your Gmail and Google Calendar, reasons about what needs to happen, calls your phone to ask — and when you say *"reply that I'll submit it by evening"* — it sends the email with those exact words. **No app. No buttons. Just talk.**

9 · Setup in 5 Minutes

```
# 1. Install dependencies
pip install -r requirements.txt

# 2. Set up environment (minimum config)
cp .env.example .env
# Edit .env — set OPENROUTER_API_KEY at minimum
# For voice: add TWILIO_ACCOUNT_SID, TWILIO_AUTH_TOKEN,
#            TWILIO_FROM_NUMBER, TWILIO_TO_NUMBER, TWILIO_WEBHOOK_BASE_URL

# 3. Google OAuth (one-time)
python scripts/setup_google_auth.py

# 4. Start Redis
redis-server

# 5. Run pre-flight check
python scripts/check.py

# 6. Launch everything
python main.py

# Open http://localhost:8080
```

For voice approval, also run `ngrok http 8000` and set the printed URL as `TWILIO_WEBHOOK_BASE_URL` in `.env`, then restart.

10 · API & MCP Reference

MCP Tools (port 8000)

TOOL	ARGUMENTS	RETURNS
`read_emails`	max_results, query	[{id, from, subject, snippet, date, unread}]
`send_email`	to, subject, body	{message_id, status}
`read_calendar`	days_ahead	[{id, summary, start, end, attendees}]
`create_event`	summary, start_datetime, end_datetime, …	{event_id, html_link, status}

`list_files`	directory	[{name, path, size_bytes, modified_iso}]
`move_file`	source, destination	{source, destination, success}

Dashboard REST API (port 8080)

ENDPOINT	METHOD	DESCRIPTION
/api/emails	GET	All plans, newest first
/api/pending	GET	Plans awaiting human approval
/api/approve/{id}	POST	Approve a plan → re-queues to Executor
/api/reject/{id}	POST	Reject and discard, logs to ChromaDB
/api/health	GET	Multi-service health (Redis, LLM, Google, Twilio, ChromaDB, MCP)
/api/metrics	GET	Confidence histogram, priority/response breakdown
/api/preferences	GET/POST	View/add ChromaDB user preference statements
/api/events/inject	POST	Inject synthetic event (demo mode)
/api/twilio/test	POST	Trigger real or simulated Twilio test call
/api/twilio/voice/{id}	GET/POST	TwiML for voice approval prompt
/api/twilio/speech/{id}	POST	Twilio SpeechResult callback → LLM intent classification
/ws	WS	WebSocket — server pushes {"type":"refresh"} on any plan change

PersonalOS Agent · SOLARIS X Hackathon 2026 · Built with Python · FastAPI · Redis · ChromaDB · Twilio · MCP · OpenRouter